

Empirical Proof Record: VALIDATED

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Proof ID: b6a1a1cc2e4fad17fe8faa1a5053985f...

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1. Claim

Across up to 4000 real FRED numbers within ± 10000 , the live number sense emits a monotone number line: $\text{Spearman}(\text{value}, \text{p_thermo}) \geq 0.90$.

- **Operationalisation:** Distinct real FRED values within range; live ArachneProtocol $\text{assign}(\text{str}(v)) \rightarrow \text{p_thermo}$; $\text{Spearman}(\text{value}, \text{p_thermo})$; distinctness; neighbour-locality (adjacent-rank prime gap vs random-pair gap); $\text{zero} \rightarrow \text{assign}('0').\text{p_thermo}$.
- **Threshold:** $\text{number_line_spearman} \geq 0.9$ rank-correlation
- **H0:** $\text{H0: Spearman} < 0.90$ — the number sense does not preserve magnitude on real data
- **H1:** $\text{H1: Spearman} \geq 0.90$ — a real magnitude number line

2. Verdict

VALIDATED — observed 0.9866205326012835

Spearman 0.9866 over 4000 real numbers $[-18.28, 9953.75]$; distinct 0.859; neighbour-locality adj/random 0.0216; zero→1409

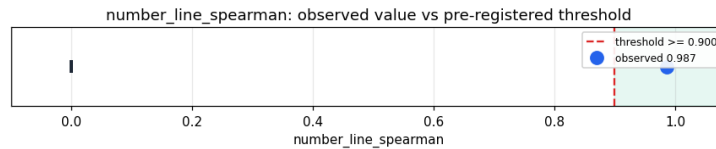


Figure 1: number_line_spearman – observed value with Wilson 95% CI.

3. Evidence

Pillar	Statistic	Value	95% CI	Library
magnitude_number_line	number_line_spearman	0.9866205326012835	(0.0000, 0.0000)	numpy 2.4

4. Pre-registration

- Registered at: 2026-05-24T03:25:04.551146+00:00
- Config hash: 0627a9d609c326ef3360bd0166fc9466...
- Data hash: cd357ab3435032f354d26ebe466929fe...

5. Signature

4148b5cfe2b58e4c9fd0c1d63980478bae3e23f4b0042ac234831918afc3afed